Ch 7: Coordinate Geometry

Question Bank | 2M | 3M | 5M

<u>SECTION B - 3 MARKS</u>

Q1. Two points A and B are on the same side of a tower and in the same straight line with its base. The angles of depression of these points from the top of the tower are 60° and 45° respectively. If the height of the tower is 15 m, find the distance between points A and B. (PYQ 2017)

Q2. The angle of elevation of the top of a tower from two points at distances a and b from the base on the same line are complementary. Prove that the height of the tower is $\sqrt{(ab)}$. (PYQ 2019 — most repeated)

Q3. From the top of a building 60 m high, the angles of depression of the top and bottom of a tower are observed to be 30° and 60° respectively. Find the height of the tower and the distance between the building and the tower. (PYQ 2019)

Q4. From a point on the ground, the angle of elevation of the top of a tower is 30° and that of the top of a flag hoisted on the tower is 60°. If the length of the flag is 5 m, find the height of the tower. (PYQ 2020)

Q5. A man standing on the deck of a ship, which is 10 m above water level, observes the angle of elevation of the top of a hill as 60° and the angle of depression of the base of the hill as 30°. Calculate the distance of the hill from the ship and the height of the hill. (PYQ NCERT)

Q6. From the top of a tower 50 m high, the angles of depression of the top and bottom of a pole are 30° and 45° respectively. Find the height of the pole and the distance between the tower and the pole. (PYQ 2020)

Q7. The angle of elevation of an aeroplane from a point on the ground is 60°. After a flight of 30 seconds the angle of elevation becomes 30°. If the aeroplane is flying at a constant height of $3000\sqrt{3}$ m, find the speed of the aeroplane. (PYQ 2019)

Q8. An aeroplane when flying at a height of 4000 m from the ground passes vertically above another aeroplane at an instant when the angles of elevation of the two aeroplanes from the same point on the ground are 60° and 45°. Find the vertical distance between the two aeroplanes at that instant. (PYQ NCERT)

Q9. A boy standing on a horizontal plane finds a bird flying at a distance of 100 m from him at an angle of elevation of 30°. A girl standing on the roof of 20 m high building finds the angle of elevation of the same bird to be 45°. Both the boy and the girl are on opposite sides of the bird. Find the distance of the bird from the girl. (PYQ 2019)

Q10. Two ships are sailing in the sea on either side of a lighthouse. The angles of depression of two ships as observed from the top of the lighthouse are 60° and 45° respectively. If the distance between the ships is $200(\sqrt{3} + 1)/\sqrt{3}$ m, find the height of the lighthouse. (PYQ)

Q11. As observed from the top of a 75 m high lighthouse from the sea level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. (PYQ 2022)

Q12. The angle of elevation of a cloud from a point 60 m above a lake is 30° and the angle of depression of the reflection of the cloud in the lake is 60°. Find the height of the cloud from the surface of the lake. (PYQ 2019)

Q13. At a point A, 20 m above the level of water in a lake, the angle of elevation of a cloud is 30°. The angle of depression of the reflection of the cloud in the lake, at A is 60°. Find the distance of the cloud from A. (PYQ NCERT)

Q14. From a window 15 m high above the ground in a street, the angles of elevation and depression of the top and the foot of another house on the opposite side of the street are 30° and 45° respectively. Find the height of the opposite house. (PYQ 2020)

Q15. A 7 m long flagstaff is fixed on the top of a tower. From a point on the ground, the angles of elevation of the top and bottom of the flagstaff are 60° and 45° respectively. Find the height of the tower correct to one decimal place. (PYQ 2019)

Q16. The angle of elevation of the top of a tower from a point A on the ground is 30°. On moving a distance of 20 m towards the foot of the tower to a point B, the angle of elevation increases to 60°. Find the height of the tower and the distance of the tower from point A. (PYQ 2020 — most repeated)

Q17. Two poles of heights 6 m and 11 m stand on a plane ground. If the distance between the feet of the poles is 12 m, find the distance between their tops. (PYQ)

Q18. From the top of a 120 m high tower, a man observes two cars on the opposite sides of the tower and in a straight line with the base of the tower with angles of depression 60° and 45° respectively. Find the distance between the two cars. (PYQ 2023)

Q19. The shadow of a vertical tower on a level ground increases by 10 m when the altitude of the sun changes from 45° to 30°. Find the height of the tower, using $\sqrt{3} = 1.73$. (PYQ 2022)

Q20. Case Study: An observer stands on the top of a 30 m high lighthouse. He observes a boat approaching the base of the lighthouse. The angle of depression of the boat changes from 30° to 60° as it moves towards the lighthouse. Find: (i) the initial distance of the boat from the base (ii) the distance moved by the boat towards the lighthouse (iii) the speed of the boat if it covers this distance in 2 minutes. (PYQ case-based 2023)

Q21. A man on the top of a vertical tower observes a car moving at a uniform speed coming directly towards it. If it takes 12 seconds for the angle of depression to change from 30° to 45°, how long will the car take to reach the tower? (PYQ 2020)

Q22. Two stations are on either side of a hill. The angles of depression of two stations from the top of the hill are 30° and 45°. If the distance between the two stations is 100 m, find the height of the hill and the horizontal distance of each station from the foot of the hill. (PYQ)

Q23. The angle of elevation of the top of a building from the foot of a tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the tower is 50 m high, find the height of the building. Also find the distance between the tower and the building. (PYQ 2019)

Q24. Two ships A and B are sailing straight away from a cliff PQ of height h. The angles of depression of two ships from the top P of the cliff are α and β respectively. Show that the distance between the ships AB = $h(\tan\alpha - \tan\beta)/(\tan\alpha \times \tan\beta)$. (PYQ HOTS)

Q25. Case Study: A school organises a trip to a nature camp. Students observe a tall pine tree from two positions X and Y on the same side of the tree. From X, the angle of elevation of the top of the tree is 60°. From Y, which is 20 m closer to the tree, the angle of elevation is 45°. The height of the tree above the eye level is h metres. Find: (i) h (ii) the distance XY (iii) the distance of point Y from the base of the tree. (PYQ 2024 case-based)

<u>SECTION C - 5 MARKS</u>

Q1. Two points A and B are on the same side of a tower and in the same straight line with its base. The angles of depression of these points from the top of the tower are 60° and 45° respectively. If the height of the tower is 15 m, find the distance between points A and B. ***(PYQ 2017)***

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